Understanding Analysis, 2nd Edition

Chapter 7 - Section 7.3 - Integrating Functions with Discontinuities

1. Straightforward.

2. Recall the Thomae's function

$$t(x) = \begin{cases} 1 & \text{if } x = 0 \\ 1/n & \text{if } x = m/n \in \mathbf{Q}\backslash\{0\} \text{ is in lowest terms with } n > 0 \\ 0 & \text{if } x \notin \mathbf{Q} \end{cases}$$

has a countable set of discontinuities occurring at precisely every rational number. Follow these steps to prove that $t(x)$ is integrable on $[0, 1]$ with $\int_0^1 t = 0$.

   (a) First argue that $L(t, P) = 0$ for any partition P of $[0, 1]$.

   (b) Let $\epsilon > 0$ and consider the set of points $D_{\epsilon/2} = \{x \in [0, 1] : t(x) \geq \epsilon/2\}$. How big is $D_{\epsilon/2}$?

   (c) To complete the argument, explain how to construct a partition P_ϵ of $[0, 1]$ so that $U(t, P_\epsilon) < \epsilon$.

   (a) By the density of the irrationals, every interval in P contains an irrational number, which makes the infimum in the interval 0.

   (b) It's finite. Let n be the largest natural number such that $\frac{1}{n} \geq \frac{\epsilon}{2}$. Then $D_{\epsilon/2}$ consists of x such that $t(x) = \frac{1}{k}$ with $k \leq n$. There are only finitely many k and for each k, there are only finitely many such x.

   A concrete example, suppose $k = 4$. How many x are there such that $t(x) = \frac{1}{4}$? There are only 2, $x = \frac{1}{4}$ and $x = \frac{3}{4}$. For $x = \frac{4}{4}$, $t(x) = 1$ and once the numerator goes past 4, we're already outside the $[0, 1]$ domain.

   In general, the number of x such that $t(x) = \frac{1}{k}$ is at most $k + 1$ because there are only $k + 1$ candidates in $[0, 1]$: $\frac{0}{k}$ up to $\frac{k}{k}$.

   (c) By part (b), there's only a finite set of rationals $R = \{r_1 = 0 < r_2 < ... < r_n = 1\}$ where $t(r_i) \geq \frac{\epsilon}{2}$. Also, since $t(x) \leq 1$ for all x, then $\frac{\epsilon}{2} \leq t(r_i) \leq 1$. The idea is to bound this finite set of large points in a small interval.

To control the interval size around the r_i's, we need to insert two points between each of them. Let $P = \{r_1, p_1, p_2, r_2, p_3, p_4, r_3, ..., r_{n-1}, p_{2n-3}, p_{2n-2}, r_n\}$. There are $2n-2$ intervals that contain r_i. Set the distance between adjacent r and p to be $< \frac{\epsilon}{4n}$. Remembering that $t(x) \leq 1$ for all x, let s_1 be the sum of all intervals that contain r_i, then

$$s_1 < (2n-2)\frac{\epsilon}{4n} < 2n\frac{\epsilon}{4n} = \frac{\epsilon}{2}$$

Now we need to consider the sum between adjacent p's. Since R contains *all* points where $t(x) \geq \frac{\epsilon}{2}$, then for $x \in [p_i, p_{i+1}]$, $x \notin R$ hence $t(x) < \frac{\epsilon}{2}$. The sum s_2 between adjacent p_i's is then

$$s_2 \leq \sum_{i=1}^{n-1} \frac{\epsilon}{2}(p_{2i} - p_{2i-1}) = \frac{\epsilon}{2}\sum_{i=1}^{n-1}(p_{2i} - p_{2i-1})$$

(we use $\leq$ because the supremum can still be $\frac{\epsilon}{2}$)

The tricky part is that we have no control over the length of individual $(p_{2i} - p_{2i-1})$. But note that since the domain is $[0, 1]$, we can conclude that the sum of interval lengths is at most 1 i.e. $\sum_{i=1}^{n-1}(p_{2i} - p_{2i-1}) \leq 1$. Then

$$s_2 \leq \frac{\epsilon}{2}\sum_{i=1}^{n-1}(p_{2i} - p_{2i-1}) \leq \frac{\epsilon}{2}$$

Finally

$$U(t, P) = s_1 + s_2 < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

3. Let

$$f(x) = \begin{cases} 1 & \text{if } x = 1/n \text{ for some } n \in \mathbf{N} \\ 0 & \text{otherwise} \end{cases}$$

Show that f is integrable on $[0, 1]$ and compute $\int_0^1 f$.

We'll use Theorem 7.3.2: prove that f is integrable on $[a, 1]$ for any $a \in (0, 1)$, then f is integrable on $[0, 1]$.

First note that $L(f, P) = 0$ for any P by density of irrationals. Now we only need to prove that $U(f, P)$ converges to 0.

There is only a finite amount of points $\{\frac{1}{k_1}, \frac{1}{k_2}, ..., \frac{1}{k_n}\}$ such that $\frac{1}{k_i} \geq a$. Create

a partition P such that each $\frac{1}{k_i}$ is contained in an interval of length less than $\frac{\epsilon}{2n}$. The upper sum of the intervals that don't contain $\frac{1}{k_i}$ is 0 because $f(x) = 0$ at those points. The total upper sum is then determined by the (at most) $2n$ intervals that contain $\frac{1}{k_i}$

$$U(f, P) < 2n\frac{\epsilon}{2n} = \epsilon$$

So $U(f) = L(f) = 0$ and $\int_0^1 f = 0$.

4. Let f and g be functions defined on (possibly different) closed intervals, and assume the range of f is contained in the domain of g so that the composition $g \circ f$ is properly defined.

   (a) Show, by example, that it is not the case that if f and g are integrable, then $g \circ f$ is integrable.

   Now decide on the validity of each of the following conjectures, supplying a proof or counterexample as appropriate.

   (b) If f is increasing and g is integrable, then $g \circ f$ is integrable.

   (c) If f is integrable and g is increasing, then $g \circ f$ is integrable.

(a) Let f be the Thomae function on $[0, 1]$ and define g

$$g(x) = \begin{cases} 0 & \text{if } x = 0 \\ 1 & \text{if } x \in (0, 1] \end{cases}$$

f is integrable by Exercise 7.3.2 and g is integrable because it only has a single discontinuity, but $g \circ f$ is the Dirichlet function which is not integrable.

(b) The answer seems to be no, but the counterexamples I can find on MSE seem to require Measure Theory.

(c) No. Part (a) serves as a counterexample.

5. Example or prove impossible

   (a) A sequence $(f_n) \to f$ pointwise, where each f_n has at most a finite number of discontinuities but f is not integrable.

(b) A sequence $(g_n) \to g$ uniformly where each g_n has at most a finite number of discontinuities and g is not integrable.

(c) A sequence $(h_n) \to h$ uniformly where each h_n is not integrable but h is integrable.

(a) Let $\{r_1, r_2, r_3, ...\}$ be the enumeration of the rationals. Define

$$f_n(x) = \begin{cases} 1, & x \in \{r_1, r_2, r_3, ..., r_n\} \\ 0, & \text{otherwise} \end{cases}$$

Each f_n only has a finite number of discontinuities and (f_n) converges pointwise to the Dirichlet function which is not integrable.

(b) Impossible. Because g_n has only a finite number of discontinuities, it's integrable. By exercise 7.2.5, if $(g_n) \to g$ uniformly and each g_n is integrable, then g is integrable.

(c) Define

$$h_n(x) = \begin{cases} 1/n, & x \in \mathbf{Q} \\ 0, & x \notin \mathbf{Q} \end{cases}$$

Each h_n is not integrable by the same reasoning that the Dirichlet function is not integrable, but (h_n) converges to the zero function uniformly.